You are a cybersecurity analyst working at a company that specializes in providing IT services for clients. Several customers of clients reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error “destination port unreachable” after waiting for the page to load.

You are tasked with analyzing the situation and determining which network protocol was affected during this incident. To start, you attempt to visit the website and you also receive the error “destination port unreachable.” To troubleshoot the issue, you load your network analyzer tool, tcpdump, and attempt to load the webpage again. To load the webpage, your browser sends a query to a DNS server via the UDP protocol to retrieve the IP address for the website's domain name; this is part of the DNS protocol. Your browser then uses this IP address as the destination IP for sending an HTTPS request to the web server to display the webpage The analyzer shows that when you send UDP packets to the DNS server, you receive ICMP packets containing the error message: “udp port 53 unreachable.”

```
13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 254

13:26:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:27:15.934126 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 320

13:28:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
13:28:50.022967 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 150
```

Cybersecurity Incident Report

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP

traffic log.

The UDP protocol reveals that: Port 53 was unreachable.

This is based on the results of the network analysis, which show that the ICMP echo reply

returned the error message: UDP port 53 unreachable.

The port noted in the error message is used for: port 53 is used for DNS protocol that translates internet domain names into IP addresses.

The most likely issue is: UDP port 53 is closed

Part 2: Explain your analysis of the data and provide at least one cause of the

incident.

Time incident occurred: 13:24:36.098564

Explain how the IT team became aware of the incident: The IT team became aware of it when users reported it and they captured the network traffic and received an ICMP error message that the UDP port 53 was unreachable.

Explain the actions taken by the IT department to investigate the incident: The IT team used TCPdump logs to capture network traffic and found out an ICMP error message.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): At 13:24:32;192571 the DNS server (203.0.113.2) was asked by the source device (192.51.100.15) through DNS query ID: 35084 to conduct a recursive desired search for the IPv4 address of yummyrecepiesforme.com. The packet length sent by the source IP was 24 bytes.

At 13:24:36.098564 the DNS server (203.0.113.2) replied to the source device (192.51.100.15) with an ICMP message of 254 bytes containing ICMP header and data transmission error details that the UDP port 53 was unreachable.

Note a likely cause of the incident:

There could be several causes regarding this issue. One of them can be the port 53 is closed.